

The cost and affordability of a healthy diet (CoAHD) indicators: methods and data sources

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BACKGROUND

One of the key reasons why millions of people are food insecure and malnourished around the world is because healthy diets are out of reach and unaffordable. Diet quality is a critical link between food security and nutrition. Poor diet quality can lead to different forms of malnutrition, including undernutrition and micronutrient deficiencies, as well as overweight and obesity. A healthy diet provides adequate calories and nutrients, but also includes a diverse intake of foods from different food groups.

Measuring and systematically monitoring the cost and affordability of healthy diets and making progress towards ensuring the affordability of healthy diets is of utmost importance and urgently needed. To this end, the Food and Agriculture Organization of the United Nations (FAO) committed to institutionalize the computation of the cost of a healthy diet, and the corresponding affordability indicators, and to publish updated estimates in the annual *The State of Food Security and Nutrition in the World* report, as well as provide the full data series on FAOSTAT.

In 2020, FAO began monitoring and reporting global, regional and country-level indicators on the cost and affordability of a healthy diet (CoAHD) to provide global evidence on people's economic access to nutritious foods to meet dietary needs for an active and healthy life, one of the core principles embedded in the definition of food security. This crucial information serves as a basis for national governments, international agencies, civil society and the private sector to work together towards improved access to healthy diets and toward achieving longstanding goals for food security, nutrition and health. Importantly, these indicators measure whether agrifood systems bring healthy diets within reach of the poorest people, considering foods that meet nutritional requirements at the lowest possible cost.

Since 2020, FAO in collaboration with the World Bank has systematically updated and reported on these indicators in the annual flagship report, *The State of Food Security and Nutrition in the World*, published by FAO, the International Fund for Agricultural Development (IFAD), the World Food Programme (WFP), the World Health Organization (WHO) and the United Nations Children's Fund (UNICEF). From 2022, the indicators are also reported annually in five *Regional Overviews of Food Security and Nutrition* reports, for Africa, the Near East and North Africa, Asia and the Pacific, Central Asia and Eastern Europe, and Latin America and the Caribbean.

The availability of these indicators at the global, regional and country levels sets the stage for increased accountability, using timely data on retail prices of diverse nutritious food items in all countries of the world. These can guide policy and programmes to help make healthy diets affordable for all people at all times, and to track progress towards this goal at the global, regional and country levels. In the coming years, FAO and the World Bank, in collaboration with participating countries, aim to expand the computation and reporting of these indicators to the subnational level, in order to further inform country policy and programming.

These indicators support efforts within the framework of the Sustainable Development Goals (SDGs) to end hunger, achieve food security and improved nutrition, and promote sustainable agriculture by 2030 (SDG 2). They also support the monitoring of progress towards the objective of transforming agrifood systems to

improve economic access to healthy diets and promoting “nutrition-sensitive agriculture.” For definitions of these indicators, please see the FAOSTAT CoAHD definitions and standards.¹

DATA AND METHODOLOGY

FAO and the World Bank compute indicators on the cost and affordability of a healthy diet in each country, which captures people’s economic access to the least-cost, locally available foods to meet requirements for a healthy diet, as defined by the food-based dietary guidelines (FBDGs). The indicators apply the global standard of a Healthy Diet Basket (HDB) to observed retail food consumer prices, along with income distributions, to provide an operational measure of people’s access to locally available foods in the proportions needed to conduct an active and healthy life. The following sections describe the methodology and data sources of these component elements.

FAO computes eight cost and affordability indicators: the cost of a healthy diet; the cost of six individual food groups that make up the Healthy Diet Basket (starchy staples; animal source foods; legumes, nuts and seeds; vegetables; fruits; and oils and fats); and the percentage and number of people unable to afford a healthy diet (see Annex 1 for the full list of indicators and definitions). The first and last indicators (cost and affordability of a healthy diet) are computed and updated each year, while the costs of the six food groups that make up the healthy diet are computed only in years when the World Bank International Comparison Program (ICP) publishes updated individual food-item price data.

The Healthy Diet Basket (HDB)

Given that the foods that make up a healthy diet vary by local context, countries have developed national food-based dietary guidelines (FBDGs) to reflect their specific cultural context, locally available foods and dietary customs (FAO, IFAD, UNICEF, WFP and WHO, 2020). FBDGs, however, are not available for all countries and those that are available are not all quantified, meaning they include specific food quantities. To overcome this data limitation and create a global standard of a healthy diet consistent with and reflective of the commonalities in dietary guidelines across countries, the CoAHD analysis applies the recommendations within and across food groups from ten quantified FBDGs and 30 semi-quantified food guides that represent a range of dietary recommendations articulated by countries.

This global standard, called the Healthy Diet Basket (HDB), is based on the average food group proportions and recommendations across FBDGs. The HDB food group amounts are the median amounts of each food group recommended in the ten quantified FBDGs (shown in Table 1), rounded to the nearest whole integer in units of 10 kcal, for ease of communication (Herforth *et al.*, 2022). The resulting food group proportions are verified for consistency by comparison to national food guides globally (see Table A2 in Annex 2). The results show that the HDB approximates a larger range of FBDGs than only the ten that were initially quantified and captures the commonalities across national guidelines.

The composition of the final HDB, shown in Table 2, indicates the dietary energy and equivalent gram content when reference foods for each food group are used to convert calories to grams. The HDB criterion is identified to meet a dietary energy intake target of 2 330 kcal per day and consists of locally available items from the six food groups mentioned above. Using this HDB, the distribution of food groups on a per-calorie basis was calculated (see Figure 1). These amounts represent an average food basket across FBDGs, for the purpose of monitoring the cost of a healthy diet.²

¹ <https://www.fao.org/faostat/en/#data/CAHD/>

² Although the HDB is calculated based on energy content of food items, it is shown by volume to demonstrate robustness in approximating a variety of national FBDGs. See Figure 7 in Herforth *et al.* (2022) for the composition of the HDB by volume, when the caloric content is converted into reference foods.

Table 1. Average food group amounts recommended across the ten national food-based dietary guidelines used to estimate the Healthy Diet Basket (scaled to meet a consistent dietary energy intake target of 2 330 kcal)

Country	Starchy staples		Vegetables		Fruits		Protein-rich foods*		Oils	
	kcal	grams	kcal	grams	kcal	grams	kcal	grams	kcal	grams
Argentina	1 000	278	145	482	228	362	638	446	320	36
Benin	1 216	338	133	443	167	266	462	323	352	40
China	1 238	344	117	391	169	269	568	398	237	27
India	1 009	280	123	409	110	175	809	566	279	32
Jamaica	1 162	323	117	389	130	206	630	441	291	33
Malta	1 343	373	112	375	148	234	572	400	155	18
Netherlands (Kingdom of the)	1 205	335	74	247	124	197	577	404	349	39
Oman	942	262	101	337	370	587	335	234	583	66
United States of America	1 038	288	89	297	187	297	753	527	263	30
Viet Nam	1 165	324	92	307	166	263	640	447	267	30
Mean	1 132	314	110	368	180	286	598	422	310	35
Median	1 164	323	114	382	167	264	604	419	285	32
Healthy Diet Basket (HDB)	1 160	322	110	367	160	254	600	420	300	34

Notes: the Healthy Diet Basket (HDB) is the global standard based on the average food group proportions and recommendations across national food-based dietary guidelines (FBDGs). * Protein-rich foods here combine dairy with other protein-rich foods, including meat, fish, egg, legumes, and/or nuts and seeds.

Source: Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C. & Masters, W.A. 2022. *Methods and options to monitor the cost and affordability of a healthy diet globally*. Background paper for *The State of Food Security and Nutrition in the World 2022*. FAO Agricultural Development Economics Working Paper 22-03. Rome, FAO. <https://doi.org/10.4060/cc1169en>

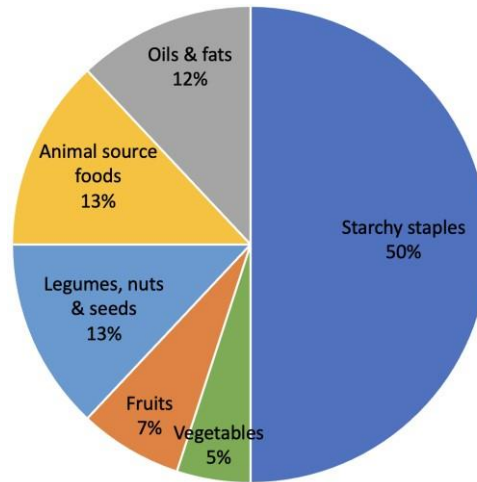
Table 2. Healthy Diet Basket content by food group, kcal and grams of reference food

Food group	Minimum number of food items selected for cost of healthy diet	Total energy content (kcal)	Equivalent gram content, by reference food (edible portion)
Starchy staples	2	1 160	322 g dry rice
Vegetables	3	110	270–400 g vegetables
Fruits	2	160	230–300 g fruits
Animal source foods	2	300	210 g egg
Legumes, nuts and seeds	1	300	85 g dry bean
Oils and fats	1	300	34 g oil

Note: To equate calories and grams for starchy staples, dry rice is the reference food; for animal source foods, egg; for legumes, nuts and seeds, dry bean is the reference food. For fruits and vegetables, the range is based on the lowest to highest kcal/g across the dataset for each.

Source: Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C. & Masters, W.A. 2022. *Methods and options to monitor the cost and affordability of a healthy diet globally*. Background paper for *The State of Food Security and Nutrition in the World 2022*. FAO Agricultural Development Economics Working Paper 22-03. Rome, FAO. <https://doi.org/10.4060/cc1169en>

Figure 1. Composition of the Healthy Diet Basket (percent of dietary energy)



Source: Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C. & Masters, W.A. 2022. *Methods and options to monitor the cost and affordability of a healthy diet globally*. Background paper for *The State of Food Security and Nutrition in the World 2022*. FAO Agricultural Development Economics Working Paper 22-03. Rome, FAO. <https://doi.org/10.4060/cc1169en>

The cost of a healthy diet

A healthy diet provides not only adequate calories but also adequate levels of all essential nutrients and food groups needed for an active and healthy life. The cost of a healthy diet is defined as the cost of purchasing the least expensive, locally available foods that meet FBDG requirements, with an energy intake of 2 330 kcal/day. The FBDGs analysed recommend food quantities for each food group and provide a wide regional representation. Although it is not selected based on nutrient content but is determined by FBDGs, this diet meets, on average, 95 percent of nutrient needs (Herforth et al., 2022; Herforth et al., 2024), and, thus, is considered nutrient-adequate.

The prices of items in each food group needed for a healthy diet were obtained using price data from the ICP as national averages for an entire year, most recently 2021 (World Bank, 2024a). Item definitions are internationally standardized, allowing classification by food group and calculation of the least-costs to reach FBDG requirements in each country, representing an average across markets and throughout the year.

The cost of a healthy diet is calculated for 173 countries in 2021 (reference year) using ICP prices to identify the lowest-cost items available in each country to meet HDB quantities of each food group. Each item is classified into its food group, as specified in Table 2, and its retail cost per day is calculated as the cost per quantity containing the energy content required for the item's food group, divided by the number of items per group. For each country, a total of 11 least-cost food items are selected in the basket: two for starchy staples, three for vegetables, two for fruits, two for animal source foods, one for legumes, nuts and seeds, and one for oils and fats. To allow substitution within energy balance, the cost per day of the least expensive locally available items in each food group is determined by the price per kcal (p_i) for each item (i) as purchased in markets, as observed in each country, multiplied by the calorie target (q_i) for that food group from Table 2, as follows:

$$\text{Cost}_{\text{StarchyStaples}} = \min (\sum_{i=1}^2 p_i q_i), \text{ where each } q_i=580 \text{ kcal } (=1\ 160/2) \text{ and item } i \text{ is a starchy staple} \quad (1)$$

$$\text{Cost}_{\text{AnimalFoods}} = \min (\sum_{i=1}^2 p_i q_i), \text{ where each } q_i=150 \text{ kcal } (=300/2), \text{ and item } i \text{ is an animal source food} \quad (2)$$

$$\text{Cost}_{\text{LegsNutsSeeds}} = \min (p_i q_i), \text{ where each } q_i=300 \text{ kcal, and item } i \text{ is a legume, nut or seed} \quad (3)$$

$$\text{Cost}_{\text{Vegetables}} = \min (\sum_{i=1}^3 p_i q_i), \text{ where each } q_i=66.7 \text{ kcal } (=110/3), \text{ and item } i \text{ is a vegetable} \quad (4)$$

$$\text{Cost}_{\text{Fruits}} = \min (\sum_{i=1}^2 p_i q_i), \text{ where each } q_i=80 \text{ kcal } (=160/2), \text{ and item } i \text{ is a fruit} \quad (5)$$

$$\text{Cost}_{\text{Oils}} = \min (p_i q_i), \text{ where each } q_i=300 \text{ kcal, and item } i \text{ is a lipid (oil or fat)} \quad (6)$$

The six food group cost indicators in equations (1) to (6) are summed to obtain the overall cost of a healthy diet per day, as follows:

$$Cost_{HealthyDiet} = \sum_{j=1}^6 Cost_j, \text{ for all } j = \{1, \dots, 6\} \text{ food groups} \quad (7)$$

The ICP database provides retail food price data for 2021, expressed in local currency units (LCU), for a wide range of internationally standardized items (716 foods and non-alcoholic beverages). These prices were reported by each country's official national statistics office (NSO) in coordination with the global office of the ICP. For international comparisons, diet costs in LCU are converted into international dollars using purchasing power parity (PPP) exchange rates for private consumption computed by the ICP and reported in the World Development Indicators (WDI) database (World Bank, 2024b). For countries with missing PPP data for three years or more, PPP imputations are applied using an autoregressive integrated moving average model with external explanatory variable (ARIMAX). For countries with missing PPP values in recent years – 43 countries in 2024 and 5 countries in 2023 – the World Bank's WDI extrapolation method is used for imputation.³

The ICP is currently the only source of retail food price data for internationally standardized items, which it collects as part of the program's larger effort to compute purchasing power parity (PPP) exchange rates across all countries. However, these data are only available once every five years, which does not allow for yearly global monitoring of diet costs to guide programmes and policies. In the absence of up-to-date food price data, a methodology was developed and validated for updating the diet cost indicator in the gap years between the ICP publication cycles (Bai *et al.*, 2024a). The methodology uses the more frequently updated national consumer food price index (CPI) data (published by FAO) to inflate the 2021 cost of a healthy diet to estimate the cost in years without ICP data.

The national CPI dataset tracks changes in monthly general and food CPIs at the national level with reference to a base year of 2015. Annual CPIs are computed as geometric averages of the 12 monthly CPIs within a year. The cost of a healthy diet is estimated by multiplying each country's 2021 actual cost, expressed in local currency units (LCU), with the food CPI ratio:

$$CoHD (PPP)_t = \frac{CoHD (LCU)_{2021} \times \left(\frac{FoodCPI_t}{FoodCPI_{2021}} \right)}{PPP_t} \quad (8)$$

Where $t = 2017, 2018, 2019, 2020, 2022, 2023$ and 2024 represents each year when the cost of a healthy diet is updated using each country's food CPI relative to the base year 2021.⁴ The cost of a healthy diet is first updated in LCU and then converted into international dollars using the WDI PPP for private consumption in each year, to compare the cost across countries and political entities.

A limitation of this method is that aggregate food CPIs do not reflect item-specific changes in food prices, nor any differential changes in the price of different food groups, due to the lack of item-level food price data over time. Currently, 170 governments report overall food CPIs while 58 countries also report more disaggregated indexes grouped by different types of foods, and only 49 countries report price level information for at least some individual food items (Bai *et al.*, 2021). Therefore, while the overall cost of a healthy diet can be updated for years without ICP data, the cost of the six food groups of a healthy diet shown in equations (1) to (6) cannot be updated due to lack of disaggregated food CPI data. As part of a medium-term strategy, FAO and the World Bank envision a partnership to expand the capture of country-level nutritious food-item price data in the gap years between the ICP publication cycles, to support updating the cost of a healthy diet indicator.

³ World Bank. 2025. How do you extrapolate the PPP conversion factors estimated by the ICP? In: World Bank Data Help Desk. [Cited 12 May 2025]. <https://datahelpdesk.worldbank.org/knowledgebase/articles/665452-how-do-you-extrapolate-the-ppp-conversion-factors>

⁴ To update the cost of a healthy diet in 2017–2020 and in 2022, 2023, and 2024, food CPI are used for all countries except the Central African Republic and Guyana, for which the general CPI is used.

The affordability of a healthy diet

To determine whether a diet is affordable or not, the cost of a healthy diet is compared with income available for food (Bai *et al.*, 2024b). To find this income line, we subtract the cost of non-food basic needs from total income. The cost of non-food basic needs is determined by finding the average expenditure shares on food and non-food of people near the poverty line, derived from household survey data from 71 countries, and then multiplying the non-food proportion by the applicable poverty lines. We apply the World Bank international poverty lines of USD 2.15 for low-income countries (LIC), USD 3.65 for lower-middle-income countries (LMIC), USD 6.85 for upper-middle-income countries (UMIC), and USD 24.36 for high-income countries (HIC). To find the expenditure shares of people near the poverty line, we use average expenditure shares from the second-lowest income quintile in LIC and LMIC, and the lowest income quintile in UMIC and HIC. When the CoHD exceeds the income available for food, accounting for the cost of non-food basic needs, it is unaffordable.

Therefore, for country i in income group j :

$$IncomeThreshold_i = CoHD_i + (IntlPovLine_j \times NonfoodExpShareLowIncome_j) \quad (9)$$

Note that we use three income classification groups with different non-food expenditure shares: low-income countries, lower-middle income, and a combined group of upper-middle and high-income countries, because the World Bank data on high-income countries includes only four nationally representative surveys (Bai *et al.*, 2024b).

The percentage of people unable to afford a healthy diet is calculated using income distributions available in the Poverty and Inequality Platform (PIP) of the World Bank (2024c). These proportions are then multiplied by population data taken from the *World Population Prospects* (UN DESA, 2022) to obtain the number of people unable to afford a healthy diet in a given country. Thus, two affordability indicators are estimated: the “prevalence of unaffordability” (PUA) and the “number of people unable to afford a healthy diet” (NUA).

For regional and global aggregates, the prevalence of unaffordability is computed as the population-weighted average of the PUA estimated for the countries for which data are available. A regional aggregate is produced only if the countries for which an estimate is available cover at least 50 percent of the aggregate’s population. The NUA_a for a given aggregate a is obtained by multiplying the average PUA_a – calculated from countries with available data – by the total population size (N_a) of all countries belonging to that aggregate. The global NUA estimate is obtained by multiplying the PUA for each of the five world regions by the total population size in each region. Calculating the global NUA estimate as the sum of the NUA estimates of other country groupings, such as those based on income levels, should be avoided.

CONCLUSION

FAO, in collaboration with the World Bank, systematically monitors and reports on food cost and affordability of a healthy diet indicators semiannually. These updates are provided in July, coinciding with the release of *The State of Food Security and Nutrition in the World* report, and in December to incorporate any newly revised data sources. In addition, periodic revisions to the entire data series are carried out to refine and improve the accuracy of the indicators as new data become available and as methodologies advance (a common practice for other FAO food security indicators). This process usually implies backward revisions of the entire series; therefore, readers are advised to refrain from comparing series across different editions of *The State of Food Security and Nutrition in the World* report and should always refer to the latest edition of the report and the data published on FAOSTAT and the World Bank, including for values in past years.

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REFERENCES

- Bai, Y., Costlow, L., Ebel, A., Laves, S., Ueda, Y., Volin, N., Zamek, M., Herforth, A. & Masters, W.A. 2021. Review: Retail consumer price data reveal gaps and opportunities to monitor food systems for nutrition. *Food Policy*, 104:102148. <https://doi.org/10.1016/j.foodpol.2021.102148>
- Bai, Y., Conti, V., Herforth, A., Cafiero, C., Ebel, A., Rissanen, M.O., Masters, W.A & Rosero Moncayo, J. 2024a. *Methods for monitoring the cost of a healthy diet based on price data from the International Comparison Program*. FAO Statistics Division Working Paper Series, No. 24-43. Rome, FAO. <https://doi.org/10.4060/cd3037en>
- Bai, Y., Herforth, A., Cafiero C., Conti V., Rissanen, M.O., Masters, W.A & Rosero Moncayo, J. 2024b. *Methods for monitoring the affordability of a healthy diet*. FAO Statistics Division Working Paper Series, No. 24-44. Rome, FAO. <https://doi.org/10.4060/cd3703en>
- FAO (Food and Agriculture Organization of the United Nations), IFAD (International Fund for Agricultural Development), UNICEF (United Nations Children's Fund), WFP (World Food Programme) & WHO (World Health Organization). 2020. *The State of Food Security and Nutrition in the World 2020. Transforming food systems for affordable healthy diets*. Rome, FAO. <https://doi.org/10.4060/ca9692en>
- FAO, IFAD, UNICEF, WFP & WHO. 2022. *The State of Food Security and Nutrition in the World 2022. Repurposing food and agricultural policies to make healthy diets more affordable*. Rome, FAO. <https://doi.org/10.4060/cc0639en>
- Herforth, A.W., Bai, Y., Venkat, A. & Masters, W.A. (forthcoming). *Healthy Diet Basket: A global standard for comparing the cost of healthy and sustainable diets*. (under review)
- Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C. & Masters, W.A. 2022. *Methods and options to monitor the cost and affordability of a healthy diet globally. Background paper for The State of Food Security and Nutrition in the World 2022*. FAO Agricultural Development Economics Working Paper 22-03. Rome, FAO. <https://doi.org/10.4060/cc1169en>
- Herforth, A., Bai, Y., Venkat, A., Mahrt, K., Ebel, A. & Masters, W.A. 2020. *Cost and affordability of healthy diets across and within countries. Background paper for The State of Food Security and Nutrition in the World 2020*. FAO Agricultural Development Economics Technical Study No. 9. Rome, FAO. <https://doi.org/10.4060/cb2431en>
- INDDX Project. 2018. Data4Diets: building blocks for diet-related food security analysis. In: *International Dietary Data Expansion Project*. [Cited 26 May 2020]. <https://inddex.nutrition.tufts.edu/data4diets>
- Lele, U., Masters, W.A., Kinabo, J., Meenakshi, J. V, Ramaswami, B., Tagwireyi, J., Bell, W.F.L. & Goswami, S. 2016. *Measuring food and nutrition security: An independent technical assessment and user's guide for existing indicators*. Measuring Food and Nutrition Security Technical Working Group. Rome, Food Security Information Network. <https://reliefweb.int/report/world/measuring-food-and-nutrition-securityindependent-technical-assessment-and-users-guide>

- Molledo, A., Troubat, N., Lokshin, M. & Sajaia, Z.** 2014. *Analyzing food security using household survey data*. Washington, DC, World Bank.
- Smith, L.C. & Subandoro, A.** 2007. *Measuring food security using household expenditure surveys*. Washington, DC, IFPRI. <https://ebrary.ifpri.org/cdm/ref/collection/p15738coll2/id/125275>
- UN DESA.** 2022. World Population Prospects 2022. In: *United Nations*. [Cited 26 September 2024]. <https://population.un.org/wpp>
- World Bank.** 2025. How do you extrapolate the PPP conversion factors estimated by the ICP? In: *World Bank Data Help Desk*. [Cited 12 May 2025]. <https://datahelpdesk.worldbank.org/knowledgebase/articles/665452-how-do-you-extrapolate-the-ppp-conversion-factors>
- World Bank.** 2024a. *International Comparison Program*. In: *World Bank*. [Cited 26 September 2024]. www.worldbank.org/en/programs/icp
- World Bank.** 2024b. DataBank: World Development Indicators. [Accessed 26 September 2024]. <https://databank.worldbank.org/source/world-development-indicators>
- World Bank.** 2024c. Poverty & Inequality Platform (PIP). In: *World Bank*. [Cited 26 September 2024]. <https://pip.worldbank.org/home>

Annex 1. Cost and affordability of a healthy diet reported on FAOSTAT

Table A1. Cost and affordability of a healthy diet indicators computed and reported on FAOSTAT

Item	Description	Year
Cost of a healthy diet Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of a healthy diet indicator is the cost of purchasing the least expensive locally available foods that meet FBDG requirements, with an energy intake of 2 330 kcal/day. The cost of a healthy diet is converted to international dollars using purchasing power parity (PPP).	2021 and updated annually
Cost of starchy staples Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available starchy staples to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Starchy staples are one of the six food groups within the Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021
Cost of animal source foods Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available animal source foods to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Animal source foods is one of the six food groups within the Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021
Cost of legumes, nuts and seeds Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available legumes, nuts and seeds to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Legumes, nuts and seeds is one of the six food groups within a Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021
Cost of vegetables Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available vegetables to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Vegetables are one of the six food groups within the Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021
Cost of fruits Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available fruits to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Fruits are one of the six food groups within the Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021

Cost of oils and fats Local Currency units (LCU) per person per day; Int\$ (PPP) per person per day)	The cost of purchasing the least expensive, locally available oils and fats to meet daily intake levels recommended in food-based dietary guidelines (FBDGs). Oils and fats are one of the six food groups within the Healthy Diet Basket (HDB). The cost of the food group is converted to international dollars using purchasing power parity (PPP).	2021
Item	Description	Year
Prevalence of unaffordability (PUA), percent	The indicator expresses the percentage of individuals in a population whose disposable income, net of the amount needed to acquire basic non-food goods and services, is lower than the average cost of the least-expensive healthy diet in a country. When CoHD exceeds the income available for food, accounting for the cost of non-food basic needs, it is unaffordable. The cost of non-food basic needs is determined by finding the average expenditure shares on food and non-food of people near the poverty line, derived from household survey data from 63 countries, and then multiplying the non-food share by the applicable poverty line. We apply the World Bank international poverty lines of USD 2.15 for LIC, USD 3.65 for LMIC, USD 6.85 for UMIC, and USD 24.63 for HIC. The derived cost of non-food basic needs is added to the CoHD to find the total income needed to afford a healthy diet. The percentage of people for whom the cost of a healthy diet is unaffordable is calculated using income distributions available in the Poverty and Inequality Platform (PIP) of the World Bank.	2021 and updated annually
Number of people unable to afford a healthy diet (NUA), million	The indicator expresses the total number of people unable to afford a healthy diet. By multiplying the PUA in a country by population data taken from the World Population Prospect, we obtain the number of people unable to afford a healthy diet in a given country in a given year.	2021 and updated annually

Source: FAO. 2025. Cost and affordability of a healthy diet – Definition and standards. In: *FAOSTAT*. Rome.
<https://www.fao.org/faostat/en/#data/CAHD>

Annex 2. Food group proportions depicted in food-based dietary guidelines

Table A2. Food group proportions (by volume) depicted in plate-shaped food guides from all countries where they are available (percent)

Country (year)	Starchy staples	Vegetables	Fruits	Vegetables and fruits	Protein-rich foods*	Oils and fats	Total
Antigua and Barbuda (2013)	39	14	14	28	28	5	100
Argentina (2015)	25	n.a.		50	20	5	100
Australia (2013)	30	28	16	44	26	0	100
Canada (2019)	25	n.a.		50	25	0	100
Chile (2013)	15	30	20	50	30	5	100
Colombia (2015)	30	n.a.		27	38	5	100
Dominica (2007)	28	25	17	42	25	5	100
Ecuador (2018)	25	n.a.		50	20	5	100
Germany (2017)	30	25	17	42	25	3	100
Grenada (2006)	47	10	10	20	27	6	100
Guyana (2018)	33	15	15	30	30	7	100
India (2018)	27	n.a.		50	20	3	100
Italy (2018)	20	27	27	54	20	6	100
Jamaica (2015)	30	24	25	49	18	3	100
Latvia (2008)	25	n.a.		50	25	0	100
Malawi (unofficial)	33	23	22	45	17	5	100
Malta (2015)	25	27	20	47	25	3	100
Mexico (2015)	33	16	16	32	35	0	100
Oman (2009)	34	15	15	30	36	0	100
Pakistan (2018)	33	19	15	34	27	6	100
Panama (2013)	38	16	16	32	25	5	100
Peru (2019)	36	16	16	32	27	5	100
Poland (2020)	25	n.a.		50	25	0	100
Portugal (2003)	27	24	20	44	26	3	100
Qatar (2015)	27	25	13	38	35	0	100
Republic of Korea (2015)	30	24	16	40	30	0	100
Switzerland (2011)	40	n.a.		40	20	0	100
United Kingdom of Great Britain and Northern Ireland (2016)	37	n.a.		37	24	2	100
United States of America (2020)	25	30	20	50	25	0	100
Uruguay (2016)	28	30	20	50	16	6	100
Zambia (2021)	34	15	18	33	33	0	100
Mean	30.1	21.7	17.6	41.0	25.9	3.0	100
Median	30.0	24.0	16.5	42.0	25.0	3.0	100
Mode	25.0	24.0	20.0	50.0	25.0	0.0	100
HDB	25.3	28.8	19.9	48.8	23.3	2.7	100

Notes: * Protein-rich foods include dairy, other animal source foods, legumes, and sometimes nuts and seeds. n.a. values are absent for fruits and vegetables as separate groups in countries where the food guide represents them in a single combined group.

Source: Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C. & Masters, W.A. 2022. *Methods and options to monitor the cost and affordability of a healthy diet globally*. Background paper for *The State of Food Security and Nutrition in the World 2022*. FAO Agricultural Development Economics Working Paper 22-03. Rome, FAO. <https://doi.org/10.4060/cc1169en>